

COMP 9200

Applied Research Methods

Carrying out a literature review

Land acknowledgment

The British Columbia Institute of Technology
acknowledges that our campuses are located
on the unceded traditional territories of the

Coast Salish Nations of

Skwxwú7mesh (Squamish),
səl'ilwətaʔt (Tsleil-Waututh), and
xwməθkwəy'əm (Musqueam)

Overview

- Why carry out a literature review?
- Finding sources for the review
- Organizing the review
- Writing the review

Literature Review – why?

- Ascertain whether others have addressed and solved your problem
- Inform you about other researchers who conduct work in same area
- Offer new ideas, perspectives, approaches that may not have occurred to you
- Alert you to controversial issues and gaps in understanding
- Bolster your confidence about the worth of your topic: others have invested considerable time, effort, resources in studying it

Literature Review – why?

- Show you how others have handled methodology/design in similar studies
- Reveal sources of data you may not have known existed
- Introduce assessment strategies others have used successfully
- Introduce you to measurement tools other researchers have developed/used
- Reveal methods of dealing with similar problem situations
- Help you interpret findings and tie results to work of those who have preceded you

Literature Review – why?

In short: the more you know about investigations and perspectives related to your topic, the more effectively you can tackle your own research problem

Literature Review - Sources

- Indexes (keywords)
- Patents
- Internet (e.g., Google scholar)
- References from previous papers

Literature Review - Sources

- Be aware of places to find literature on your topic
 - Books
 - Journals
 - Newspapers / magazines
 - Government publications / white papers
 - Conference presentations
- Identify one or more keywords
 - Based on your research problem or question and its subproblems

Literature Review - Sources

- Searches of thousands of journals and other sources
 - Including abstracts or entire documents
- Web of Science: helpful database for literature search
 - Information about which publications cite other publications
 - Citation Network feature for searching reference lists
- Electronic copies published by professional associations
- Open-access journals on the Internet (some peer-reviewed)
- See Table 3.1 from textbook

Literature Review - Sources

Librarians:

- Their purpose: help you and others find information
- Beneficial to consult early in your literature search
 - Save you time and effort in finding unfamiliar and valuable resources; reduce your frustration
 - Teach you how to use resources and access databases
- Consult face-to-face and in email or chat

Literature Review - Sources

- General strategies for accessing search engines on websites
 - Use at least two keywords to limit your search
 - If you want to look for a phrase rather than a single word, use quotation marks
- Access to government documents and publications of professional organizations
- Wikipedia: good general overviews; not peer-reviewed
 - Not to be used as an authoritative source

Literature Review - Sources

- Additional resource: literature reviews of other researchers
 - Track down references cited by three or more other researchers
- Go to the original source and read it when possible
 - Beware of authors misrepresenting the work of a particular researcher

Literature Review - Sources

- Record the following information about each source
 - Author, title, date of publication, page numbers, publication information
 - For Internet sources: U R L (uniform resource locator), date retrieved, D O I (digital object identifier)
- Adhere to an appropriate style manual
 - Different disciplines prescribe different styles
 - See Table 3.2: Commonly Used Styles in Research Reports
 - A P A, Chicago, C S E, M L A

Note: citation managers (e.g., Zotero) are helpful!

Literature Review - Sources

- When is enough?
- Always new things to learn about a topic
 - Continue to look for new research reports
- Indicators that you have searched sufficiently for your preliminary literature review
 - Repetitive patterns in the materials you are reading
 - Arguments, methodologies, and findings that become familiar
 - Same key researchers and studies cited over and over
 - No longer encountering new viewpoints

Literature Review - Organizing

- Never take other people's conclusions at face value
- Determine for yourself whether conclusions are justified based on data presented
- Organize ideas encountered during review

Literature Review - Organizing

- Organizing and synthesize into a cohesive review:
 - Compare/contrast varying theoretical positions
 - Show how approaches have changed over time
 - Describe general trends in research findings
 - Identify/explain discrepant or contradictory findings , and suggest possible explanations for such discrepancies
 - Do not intentionally bias your review by omitting or misrepresenting prior research findings
 - Identify general themes that run throughout the literature

Literature Review - Writing

- Develop a plan for the overall organizational structure of your review
- Emphasize relatedness (how literature related to the problem)
- Give credit where credit is due (use appropriate citations)
- Review literature, do not reproduce it
 - Know the difference between describing the literature and plagiarizing it
 - Minimize your use of direct quotations from other people's writings

Literature Review - Writing

- Provide transitional phrases, sentences, or paragraphs that help your readers follow your train of thought
- Summarize what you have said
- Remember that your first draft will not be your last draft
- Ask others for advice and feedback
- See the example at the end of Chapter 3

Review

- Why carry out a literature review?
- Finding sources for the review
- Organizing the review
- Writing the review